

physical machine. In this scenario, an application is not installed on it but will act like it is installed on it. All the application related information is available on the remote server.

Desktop virtualisation: In this kind of virtualisation, the users' operating system is stored on the remote server and they are allowed to use it as a virtual desktop, anywhere in the world.

User virtualisation: This is similar to desktop virtualisation, but it has the capability of maintaining a personalised virtual desktop. Users can log into their virtual desktop using different types of devices.

Storage virtualisation: In this case, different network storage devices combine and act like a single storage device.

Hardware virtualisation: This is a very popular option and forms the basis of cloud computing. Here, one processor acts like a different virtual processor and more users can access the same physical processor at the same time. This kind of virtualisation requires a virtual machine manager called a hypervisor.

The benefits of virtualisation are:

- Leads to an exponential growth in productivity
- Better resource utilisation
- Offers users agility and good efficiency
- Cost reduction due to not having to buy and maintain a large number of physical servers
- Data centres become smaller and compact, compared to using only physical servers, which provide the following benefits:
 - Energy savings
 - Less hardware maintenance
 - Considerable savings in time and money

Figure 1 demonstrates the difference between traditional and virtual architecture. As we can see, in virtual architecture, we can create pools of OSs and apps using the virtual layer.

Hypervisor

A hypervisor is software or firmware that creates virtual machines. It is also called a virtual machine monitor. Basically, it is responsible for managing virtual machines.

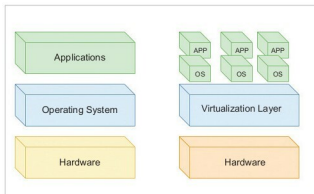


Figure 1: Traditional architecture vs virtual architecture

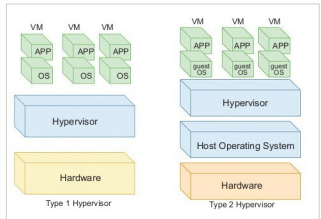


Figure 2: Type 1 hypervisor vs Type 2 hypervisor

There are two types of hypervisors: Type 1 and Type 2. Figure 2 demonstrates the difference between them.

Type 1 hypervisor: This directly runs on system hardware, and is also called the bare metal hypervisor.

Type 2 hypervisor: This runs on the host operating systems, above which VMs are created. And the OSs, which are installed inside VMs, are called guest OSs.

Virtual machines vs containers

Virtualisation has played a major role in improving system utilisation, isolation, micro services, modern computing, and much more. Virtual machines and hypervisors make up one approach towards workload deployment, while using containers is a new approach that is more efficient and reliable compared to traditional virtualisation.

The main difference between a container and a VM is in the virtual layer, or the way OS resources are assigned. In the case of VMs, the first hypervisor is normally installed on bare metal hardware, and once it's installed, VMs are created from the system's available resources. Here, each VM has its own virtual OS and application. Usually, the first VM is used as a host – it has the management software, while other VMs have different applications/services like the CRM, databases, JVM, etc.

In the case of containers, the arrangement is totally different. Here, the host OS is installed first and then the container layer (Docker engine) is installed, like LXC (a Linux container). Container instances that are created on the container layer use the host OS only – this layer doesn't have its own virtual OS and applications. So it is more efficient and a greater number of container instances can be created compared to VMs.

One of the disadvantages of the container approach is a single point of failure. If all the containers are using the same host OS, and if it is hacked or crashes, then it can create issues. Also, container migration has some limitations, such as other servers should have the same or compatible OS kernels.